

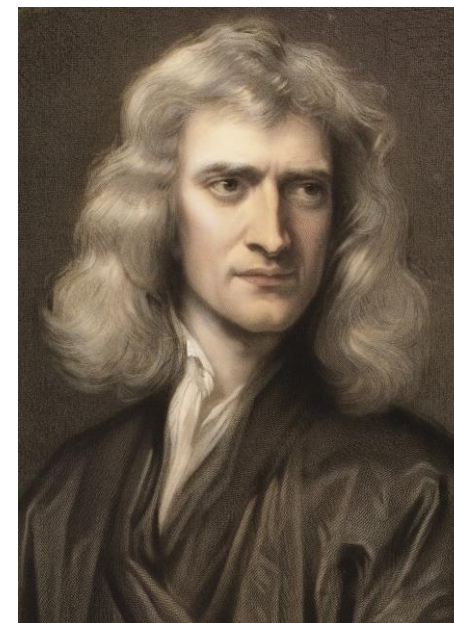
Math Primer

(6. Integration)

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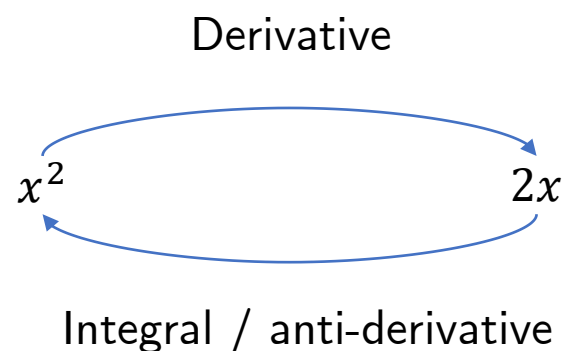
Isaac Newton
(1643-1727)



G.W. Leibniz
(1646-1716)

Integrals as anti-derivatives

- ▶ Derivative of a function $f(x)$ gives the slope of the function $f'(x)$. If x is a free input variable, the derivative itself is a net new function. Example, $f(x) = x^2$, then $f'(x) = 2x$ which is another function
- ▶ Naturally, the next question is given the derivative of a function, what is the function ? It's a reverse engineering problem and harder to calculate than derivatives.



- ▶ We will shortly see that the anti-derivative is **not unique**

Properties of Indefinite Integral

▶ When we arrive at the integral in a functional form, it's called **indefinite integral**.

▶ The notation for indefinite integral is

$$\int (.)dx$$

▶ The function that is being integrated is called **Integrand** and x is the integration variable

▶ Because the derivative of a constant is 0, we add it to the known anti-derivative to express the most general function possible.

▶ Therefore

$$\int f'(x)dx = f(x) + C$$

C is an arbitrary constant (sometimes called as constant of integration)

Properties of Indefinite Integral



Some basic Integration formulae

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int dx = x + C$$

$$\int e^x dx = e^x + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

Properties of Indefinite Integral

► Integral is a Linear operator

$$\int (\alpha f'(x) \pm \beta g'(x)) dx = \alpha f(x) \pm \beta g(x) + C$$

Exercise 1: $\int (3x^2 + 4x^3) dx$

Exercise 2: $\int \left(5e^x - \frac{4}{x} \right) dx$

Exercise 3: $\int \left(\frac{x^3 - 1}{x^2} \right) dx$

Methods of Integration – Substitution

- ▶ Most of the times, the integrand will be a composite function of x . In this case, we make use of intermediate variables (method of **substitution**) which is the reverse of chain rule.
- ▶ So, if we have $\int f(g(x))dx$ then assume $g(x) = t$, so $\frac{dt}{dx} = g'(x)$

Exercise 1 $\int e^{5x} dx$

Exercise 2 $\int (2x \cos(x^2 + 1)) dx$

Exercise 3 $\int \frac{2x}{1+x^2} dx$

Exercise 4 $\int \frac{1}{x + x \ln x} dx$

Exercise 5 $\int \frac{e^{2x} - 1}{e^{2x} + 1} dx$

Methods of Integration – Substitution



Extra references

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

Methods of Integration – Partial Fractions

- ▶ Below are some specific handling of integrands when they are rational functions. Equate coefficients on RHS with LHS for powers of x . RHS is easy to integrate.

Rational Function	Partial Fractions
$\frac{px + q}{(x - a)(x - b)}$	$\frac{A}{x - a} + \frac{B}{x - b}$
$\frac{px + q}{(x - a)^2}$	$\frac{A}{x - a} + \frac{B}{(x - a)^2}$
$\frac{px^2 + qx + r}{(x - a)(x - b)(x - c)}$	$\frac{A}{x - a} + \frac{B}{x - b} + \frac{C}{x - c}$
$\frac{px^2 + qx + r}{(x - a)^2(x - b)}$	$\frac{A}{x - a} + \frac{B}{(x - a)^2} + \frac{C}{x - b}$
$\frac{px^2 + qx + r}{(x - a)(x^2 + bx + c)}$	$\frac{A}{x - a} + \frac{Bx + C}{x^2 + bx + c}$

Methods of Integration – Partial Fractions



Exercises – integrate the following

$$\int \frac{1}{(x+1)(x+2)} dx$$

$$\int \frac{x}{(x-1)^2} dx$$

$$\int \frac{1}{x(x^2+1)} dx$$

$$\int \frac{1}{x^4-1} dx$$

Methods of Integration – By parts



- Integration of product of two functions is given by **integration by parts** formula. It's the reverse of product rule of derivative

$$\int (uv)dx = u \int vdx - \int \left[u' \int vdx \right] dx$$

How to choose u and v ? Well, there is a rule of thumb for u

It's called **ILATE** (Inverse Trig, Log , Algebraic, Trig, Exponential)

Exercise 1 $\int x \cos x \, dx$

Exercise 2 $\int \ln x \, dx$

Exercise 3 $\int x e^x \, dx$

Exercise 4 $\int e^x \sin x \, dx$

Definite Integral

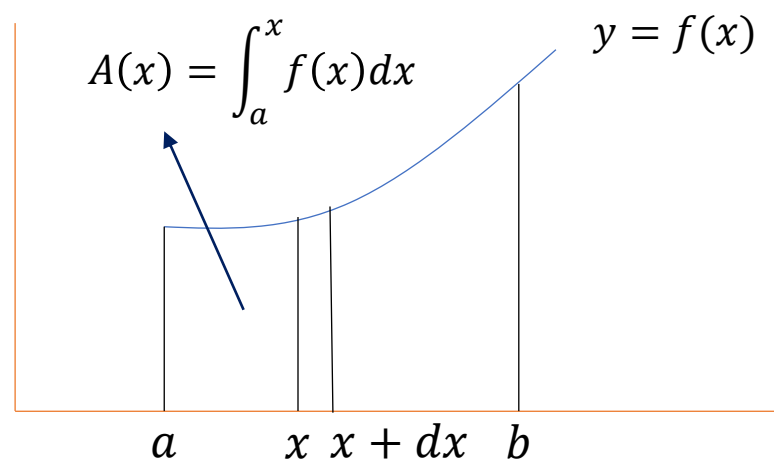


- ▶ In contrast to indefinite integral which is a function, a definite integral is a unique value. It's denoted as

$$\int_a^b f(x) dx$$

a is the lower limit and b is the upper limit

- ▶ Graphically, it's the area under the curve $f(x)$ from a to b
- ▶ The value of this integral is given as $F(b) - F(a)$, where $F(x)$ is the antiderivative of $f(x)$



Definite Integral



Exercises – integrate the following

$$\int_{-2}^2 x dx$$

$$\int_{-2}^2 x^2 dx$$

$$\int_2^4 f(x) dx, \text{ where } f(x) = \begin{cases} 2, & x < 3 \\ x - 1, & x \geq 3 \end{cases}$$

Definite Integral – With substitution



- ▶ When we perform variable transformation as part of substitution, we also need to change the limits based on the new variable used

$$\int_0^1 \frac{x}{x^2 + 1} dx$$

$$\int_0^2 x\sqrt{x+2} dx$$

Definite Integral – Properties

Some very useful formulas to calculate definite integrals

$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

$$\int_a^b f(x)dx = -\int_b^a f(x)dx \quad \int_a^a f(x)dx = 0$$

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$\int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx \quad , \text{ If } f(x) \text{ is even, else } 0$$